

1. **Experiment:** Genomic DNA third-generation sequencing

2. **Time:** 2026.01.24-2026.02.01

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4. **Method:**

(1) Genomic DNA Extraction

① Cells harvested from the final round of sorting were collected by centrifugation at $300 \times g$ for 5 min at room temperature. The supernatant was carefully removed and the cell pellet was resuspended in 200 μL of Buffer GA (provided in the TIANamp Genomic DNA Kit) and vortexed until completely suspended.

② Proteinase K (20 μL) was added to the suspension and mixed thoroughly.

③ Buffer GB (200 μL) was added and the mixture was inverted to mix completely, followed by incubation at 70 $^{\circ}\text{C}$ for 10 min. The solution became clear upon complete lysis. A brief centrifugation was performed to remove droplets from the inner surface of the tube lid.

④ An equal volume of anhydrous ethanol (200 μL) was added to the lysate and the mixture was vortexed vigorously for 15 s to ensure complete mixing, after which a brief centrifugation was performed to remove droplets from the tube lid.

⑤ The entire mixture (including any flocculent precipitate) was transferred to a Spin Column CB3 placed in a 2 mL Collection Tube, and centrifuged at 12,000 rpm for 30 s. The flow-through was discarded and the column was placed back into the collection tube.

⑥ Buffer GD (500 μL) was added to the column and centrifuged at 12,000 rpm for 30 s. The flow-through was discarded and the column was returned to the collection tube.

⑦ Buffer PW (600 μL) was added to the column and centrifuged at 12,000 rpm for 30 s. The flow-through was discarded and the column was placed back into the collection tube. Repeat this step for collecting more end-product.

⑧ The column was centrifuged at 12,000 rpm for 2 min to remove residual wash buffer, and the flow-through was discarded. The column was then placed at room temperature for 3 min to completely air-dry the membrane.

⑨ The column was transferred to a clean 1.5 mL microcentrifuge tube, and 100 μL of Buffer TE was added dropwise onto the center of the membrane. After incubation at room temperature for 3 min, the column was centrifuged at 12,000 rpm for 2 min to elute the genomic DNA. The eluted DNA was quantified using a Nanodrop spectrophotometer and stored at $-20\text{ }^{\circ}\text{C}$ for subsequent PCR amplification.

(2) Polymerase Chain Reaction (PCR)

① PCR primers were designed on the constant regions of the vector backbone, i.e., upstream of fragment A and downstream of fragment C, respectively, to amplify the complete inserted fragment.

② The PCR reaction was performed in a 20 μL volume. A negative control (nuclease-free water instead of DNA template) was included in each run. The reaction mixture composition was as follows:

Table 1. PCR reaction mixture composition (20 μL total volume).

Components	Volume
Nuclease-free water (DEPC water)	up to 20 μL
2 $\times$ Phanta Max Master Mix (Vazyme)	10 μL
Forward primer (upstream of A, 10 μM)	0.8 μL
Reverse primer (upstream of C, 10 μM)	0.8 μL
Genomic DNA template	2 μL (80 ng)

③ The PCR amplification was carried out under the following cycling conditions:

Table 2. PCR program.

Step	Temperature	Time	Cycles
Initial denaturation	95 °C	3 min	1
Denaturation	95 °C	15 s	
Annealing	60 °C	15 s	30
Extension	72 °C	30 s/kb	
Final extension	72 °C	5 min	1

- ④ The PCR products were verified by agarose gel electrophoresis to confirm the specificity of the amplified bands. The target bands were excised and purified using a gel extraction kit. The purified products were quantified using a Nanodrop spectrophotometer.
- (3) Gel purification and quantification
- ① The PCR products were verified by agarose gel electrophoresis to confirm the specificity of the amplified bands.
- ② The target bands were excised and purified using a gel extraction kit.
- ③ The purified products were quantified using a Nanodrop spectrophotometer.
- (4) PacBio third-generation sequencing
- ① The purified PCR products were submitted to a commercial sequencing service for third-generation sequencing on the PacBio platform. The sequencing data were then analyzed as follows: the reads were anchored by the homology arm sequences at both ends of each fragment. Specifically, the A/B junction (Flag tag sequence) defined the boundary between fragment A and B; the B/C junction (GGGGS linker sequence) defined the boundary between fragment B and C; and the left end of A and the right end of C defined the boundaries between the insert and the vector. Through localization of these homology arms, the exact nucleotide sequences of fragments A, B, and C were deconvoluted from the full-length sequences.

5. Result:

- (1) The PCR products showed a single specific band at the expected size on agarose gel electrophoresis, confirming successful amplification of the complete insert from the genomic DNA. No amplification was observed in the negative control.
- (2) A total of 19,606 valid sequences were obtained from PacBio sequencing, of which 14,894 were full-length sequences (containing complete fragments A, B, C and flanking homology arms), accounting for 75.97% of the total.
- (3) The sequencing reads were anchored by the homology arm sequences at both ends of each fragment: the A/B junction (Flag tag sequence) defined the boundary between fragment A and B; the B/C junction (GGGGS linker sequence) defined the boundary between fragment B and C; and the left end of A and the right end of C defined the boundaries between the insert and the vector. Through localization of these homology arms, the exact nucleotide sequences of fragments A, B, and C were deconvoluted from the full-length sequences and translated into amino acid sequences. Each clone was mapped to one of the 27 initial binder variants, and the occurrence frequency of each variant among the surviving cells was counted.
- (4) Variants were ranked by enrichment, and the top-ranking sequences were selected for downstream affinity validation.

Table 3. Sequencing results of the 27 candidate sequences after the first round of screening.

			Final Type	Total Counts
			A2B3C3	2530
			A2B3C2	1157
			A3B2C2	1012
			A1B2C2	942
			A1B3C3	919
			A1B1C2	811
			A3B1C2	782
			A3B2C3	700
			A1B1C1	657
			A1B3C1	559
			A2B1C3	553
			A2B2C2	545
			A1B1C3	472
			A2B2C1	411
			A2B1C1	396
			A3B3C3	371
			A3B3C1	317
			A3B3C2	317
			A3B1C1	297
			A3B2C1	277
			A1B2C3	270
			A1B2C1	208
			A2B3C1	150
			A1B3C2	123
			A2B1C2	118
			A2B2C3	0
			A3B1C3	0
SumTotal Counts	Seq Counts	Perfect Percentage		
14894	19606	75.9665%		

Note. Final Type, identifier of the final assembled construct, where A, B, and C denote the three segments and the numbers indicate the specific variant of each segment. Total Counts, the total number of sequencing reads obtained for each specific assembled type. Data are sorted in descending order of Total Counts. Sum Total Counts, the total number of sequencing reads corresponding to the target sequences. Seq Counts, the total number of valid sequencing reads obtained from the enriched library. Perfect Percentage, the percentage of correctly assembled sequences, calculated as (Sum Total Counts / Seq Counts) × 100%.

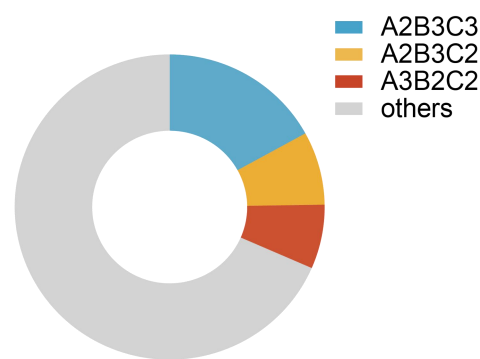


Figure 1. Distribution of full-length A-B-C combinations among 14,894 sequencing reads recovered after sorting.

6. AI Use Statement:

AI-assisted data analysis and modeling were conducted in PacBio read anchoring and variant enrichment ranking. The methods and outputs were reviewed and validated by the team.